

ROB 545: Kinematics, Dynamics, & Control

Forward kinematics

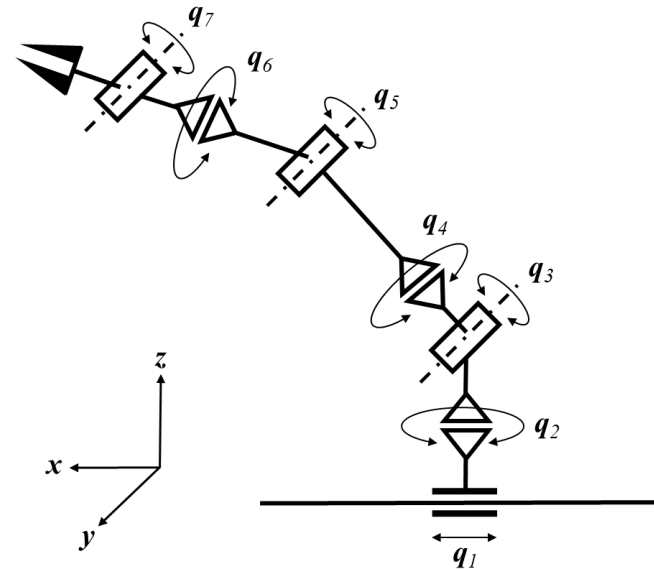
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Goals for today

- Solve the forward kinematics using the product of exponentials (PoE) formula
- Work through a few example problems

Kinematics

Kinematics is a branch of classical mechanics that describes the motion of points, bodies (objects), and systems of bodies (groups of objects) without considering the mass of each or the forces that caused the motion

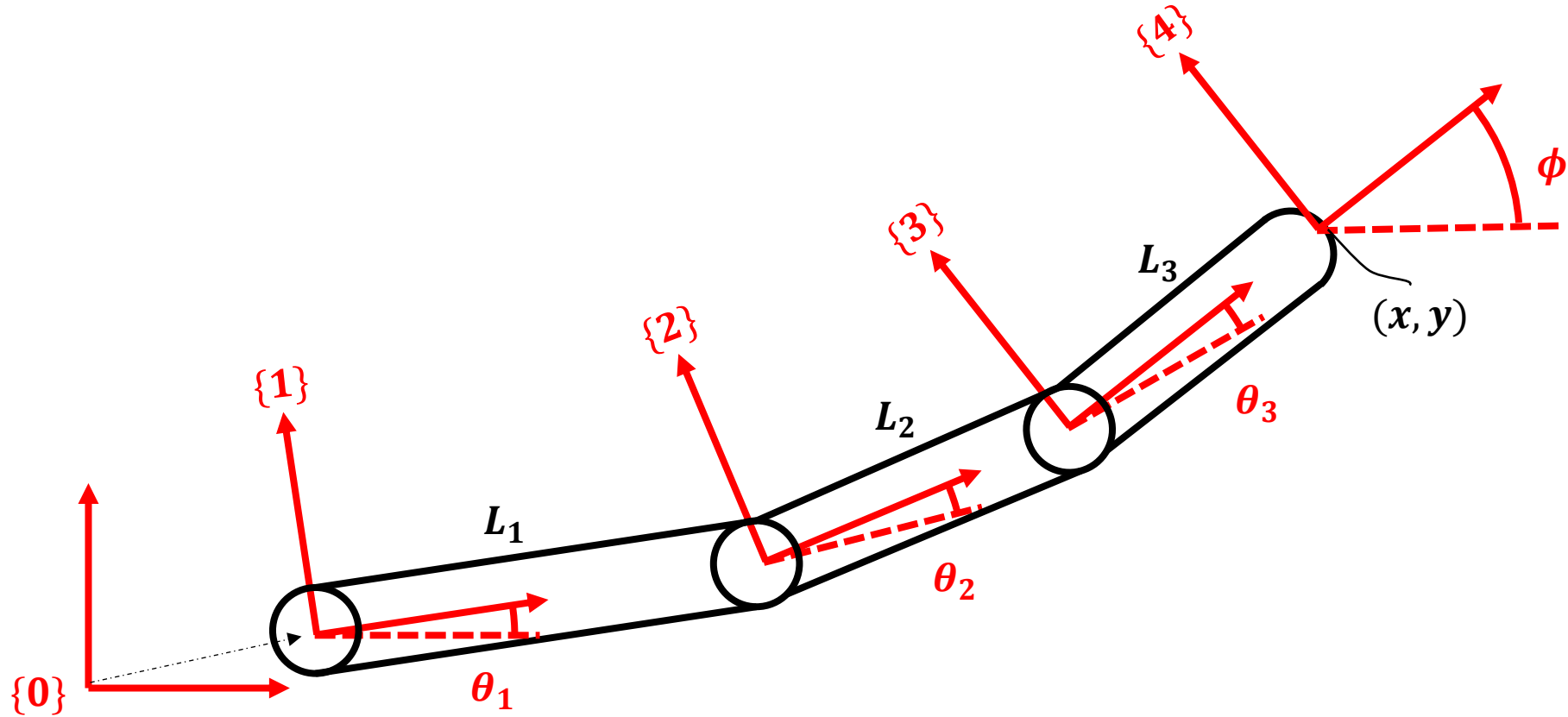


What is forward kinematics (FK)?

Forward kinematics: calculation of the position and orientation of the robot's end-effector frame from its joint coordinates

Inverse kinematics is the reverse – determining the joint coordinates from the end-effector frame's position and orientation. We will consider this problem later.

Let's start with a simple example: a 3R planar arm



We can use basic geometry for this!

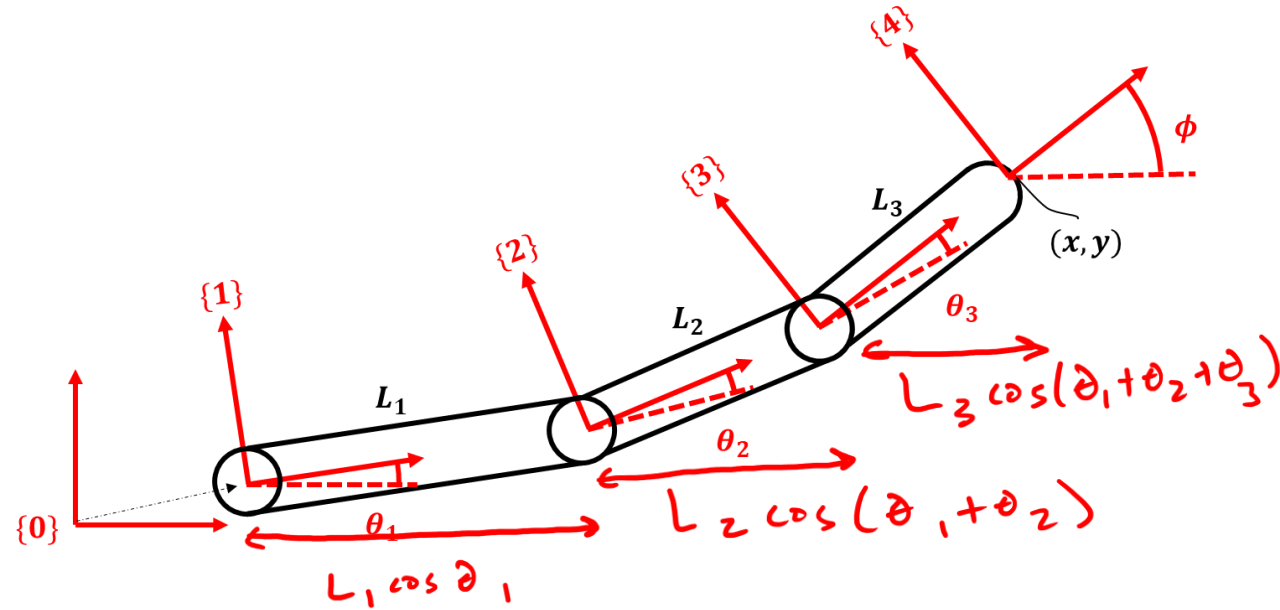
For the end-effector:

$$x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

$$\phi = \theta_1 + \theta_2 + \theta_3$$

This is a simple robot! Trigonometry would be very complex for general spatial chains.



FK with transformations

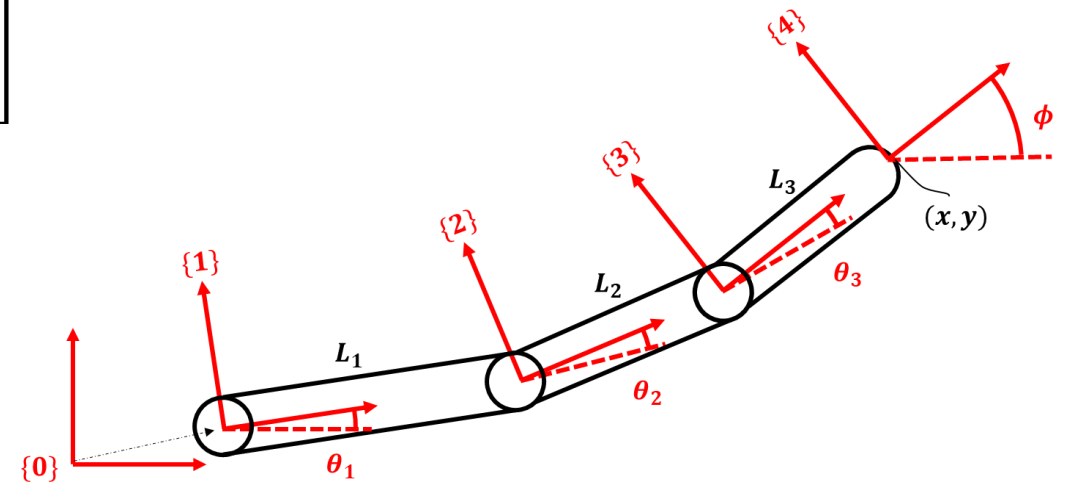
We can also express the FK as a product of 4 homogeneous transformation matrices:

$$T_{04} = T_{01}T_{12}T_{23}T_{34}$$

where

$$T_{01} = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 0 \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_{12} = \begin{bmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & L_1 \\ \sin(\theta_2) & \cos(\theta_2) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{23} = \begin{bmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & L_2 \\ \sin(\theta_3) & \cos(\theta_3) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_{34} = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Solution techniques

There are two different approaches for representing the forward kinematics of general open chains:

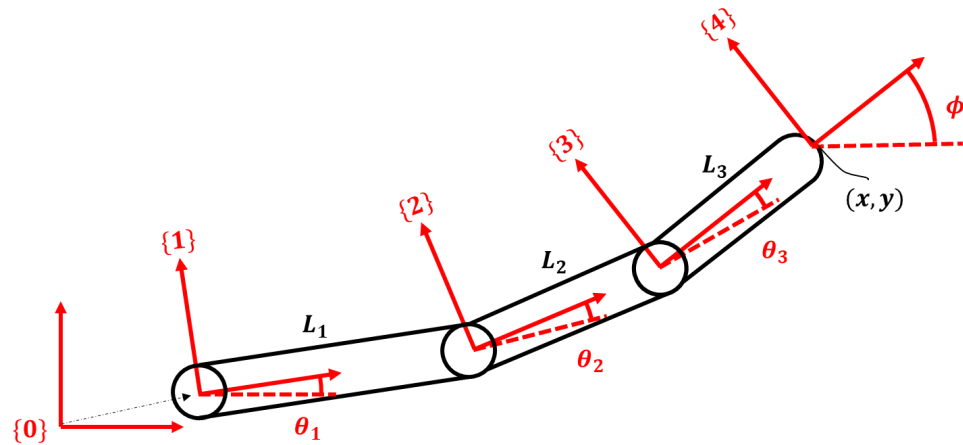
1. Denavit-Hartenberg (DH) method

- Assigns a reference frame to each link using a strict convention
- Uses a minimal number of parameters
- Most popular, but can be a bit cumbersome (see Appendix C for details)

2. Product of Exponentials (PoE) formula

- Not a minimal representation, but no link frames are necessary
- More intuitive
- **We'll use this representation**

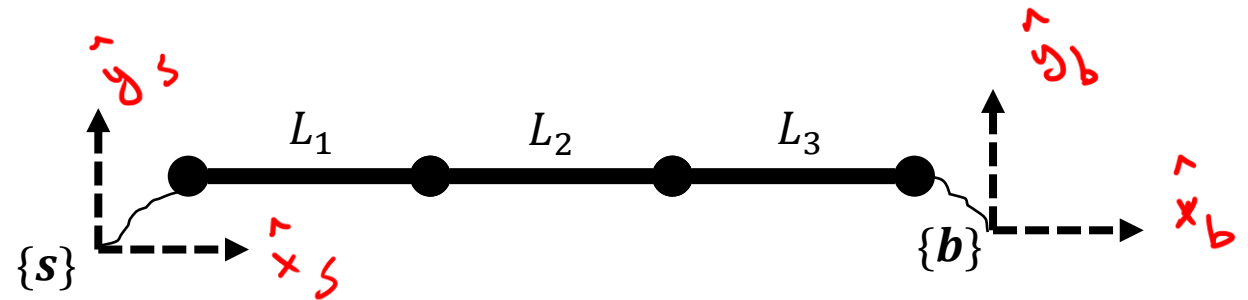
PoE formula



$$T_{sb}(0,0,0) = \underbrace{\begin{bmatrix} 1 & 0 & 0 & L_1 + L_2 + L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_M$$

Our goal is to find $T_{sb}(\theta_1, \theta_2, \theta_3) = ?$

Let's start with a simple case where $\theta_1, \theta_2, \theta_3 = 0$



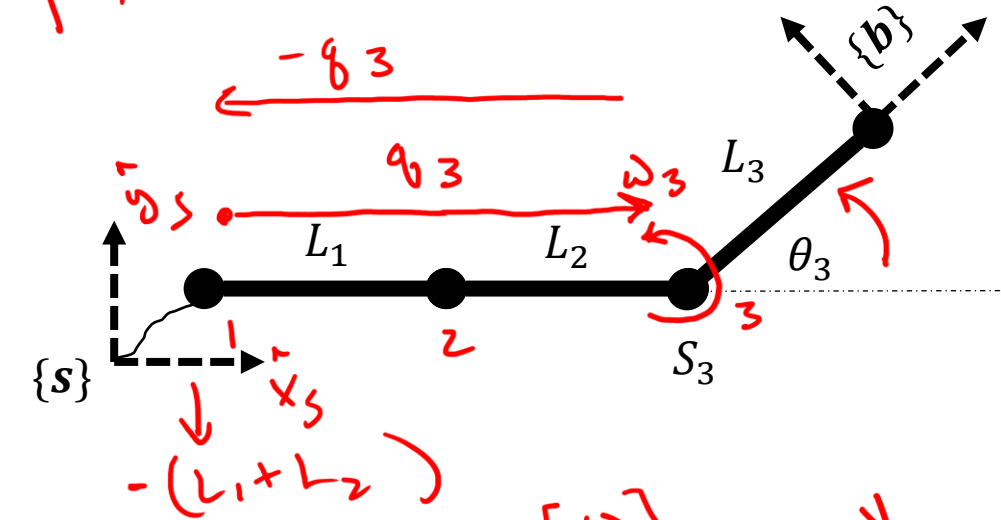
is the transformation from $\{0\}$ to $\{4\}$ when all joint angles are set to zero (also known as the "home" configuration)

PoE formula

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & -1 \\ L_1+L_2 & 0 & 0 \end{vmatrix} = \hat{i}(0) - \hat{j}(L_1+L_2) + \hat{k}(0)$$

Now rotate around joint 3 by θ_3

This is a screw motion with axis S_3 . We express the screw axis in the fixed $\{s\}$ frame ($\theta_1, \theta_2 = 0$)



In matrix form

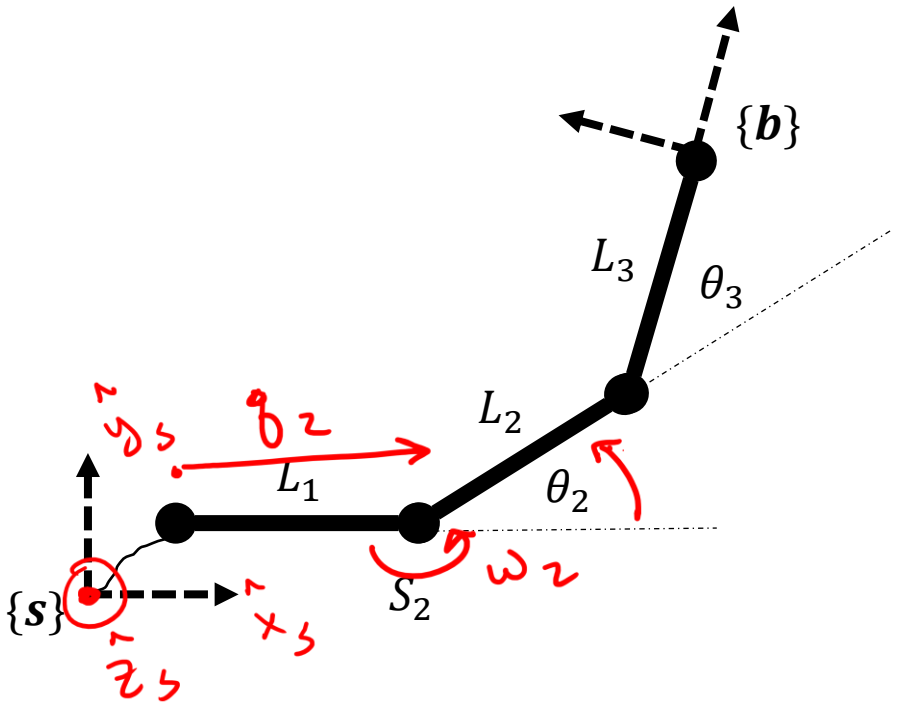
$$[S_3] = \begin{bmatrix} [\omega] & v \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}$$

$$S_3 = \begin{bmatrix} \omega_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -\omega_3 \times q_3 + 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}$$

So, for any θ_3 where $\theta_1 = \theta_2 = 0$

$$T_{04}(0,0,\theta_3) = e^{[S_3]\theta_3} M$$

PoE formula



Now rotate about joint 2 (screw axis) by θ_2 (holding $\theta_1 = 0$)

$$\mathcal{S}_2 = \begin{bmatrix} \omega_2 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -\omega_2 \times q_2 + 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -L_1 \\ 0 \end{bmatrix}$$

$$[\mathcal{S}_2] = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -L_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} T_{04}(0, \theta_2, \theta_3) &= e^{[\mathcal{S}_2]\theta_2} T_{04}(0, 0, \theta_3) \\ &= e^{[\mathcal{S}_2]\theta_2} e^{[\mathcal{S}_3]\theta_3} T_{04}(0, 0, 0) = e^{[\mathcal{S}_2]\theta_2} e^{[\mathcal{S}_3]\theta_3} M \end{aligned}$$

PoE formula

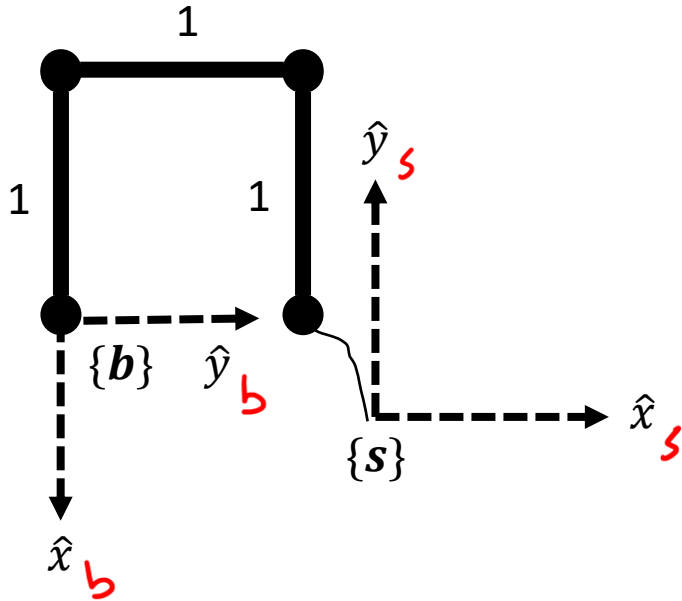
Finally, for rotation about joint 1, which applies a screw motion to the entire 3-link assembly:

$$T_{04} = e^{[S_1]\theta_1} e^{[S_2]\theta_2} e^{[S_3]\theta_3} M$$

$$[S_1] = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

*This works for any arbitrary θ_1 , θ_2 , and θ_3

Let's verify the PoE formula



Let's say that each joint angle is 90° and that each link has a length of 1. By visual inspection we see that

$$T_{sb} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

```

% Compute the FK of a planar 3R arm
L = 1; % link lengths of 1
M = [1 0 0 3*L; 0 1 0 0; 0 0 1 0; 0 0 0 1]; % home position
theta = pi/2; % joint angles of 90 deg
S3 = [0 0 1 0 -(L+L) 0]; % screw axis 3
S2 = [0 0 1 0 -L 0]; % screw axis 2
S1 = [0 0 1 0 0 0]; % screw axis 1
% PoE formula
Tsb = MatExp(S1,theta)*MatExp(S2,theta)*MatExp(S3,theta)*M

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function T = MatExp(S,theta)
% calculate the homogeneous transformation for a set of
% exponential coordinates
% Inputs: S - screw axis; theta - angle;
% Output: T - homogeneous transformation matrix
wSkew = [0, -S(3), S(2);
         S(3), 0, -S(1);
        -S(2), S(1), 0];
v = [S(4), S(5), S(6)]';
R = eye(3)+sin(theta)*wSkew+(1-cos(theta))*wSkew^2;
p = (eye(3)*theta+(1-cos(theta))*wSkew+(theta-
sin(theta))*wSkew^2)*v;
T(1:3,1:3) = R;
T(:,4) = p;
T(4,:) = [0 0 0 1];

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Command Window

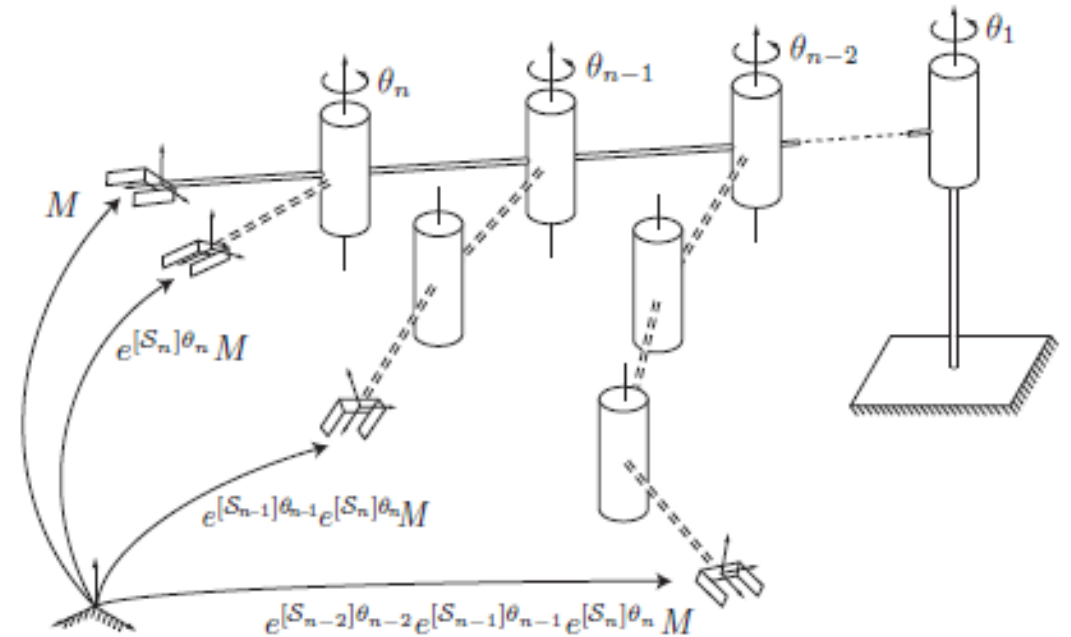
Tsb =

-0.0000	1.0000	0	-1.0000
-1.0000	-0.0000	0	0
0	0	1.0000	0
0	0	0	1.0000

Matches what we found by visual inspection!

PoE: Main Idea

1. Consider each joint as applying a screw motion to all outward links.
2. While defining a frame at each link is not required for this approach, it will be helpful for visualization and calculating link inertia (dynamics).



PoE: Steps

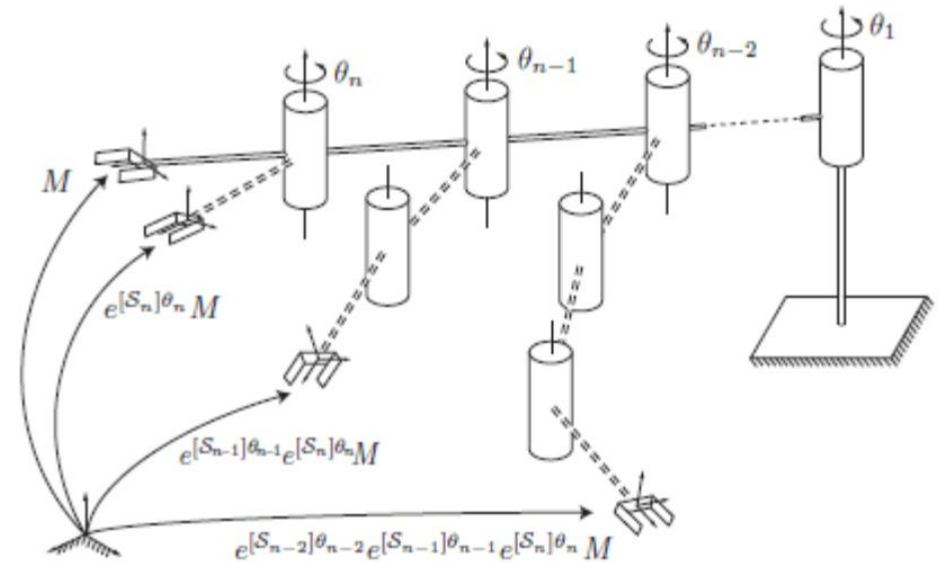
Step 1

- Choose a fixed base frame $\{s\}$ and an end-effector frame attached to the last link $\{b\}$
- Set all joint angles to 0
- Find $M \in SE(3)$, the configuration of $\{b\}$ relative to $\{s\}$ when the value of all joints are set to 0

Step 2

- Suppose joint n is displaced to some joint value θ_n . This will lead to a transformation of $\{b\}$ where $T = e^{[S_n]\theta_n}M$ and S_n is the screw axis of joint n in the $\{s\}$ frame
- Complete this for $n, n - 1, \dots, 1$ until you have

$$T_{sb}(\theta_1, \dots, \theta_n) = e^{[S_1]\theta_1} \dots e^{[S_n]\theta_n} M$$



Screw axes for revolute & prismatic joints

Revolute (rotating) joints

- $\omega_n \in \mathbb{R}^3$ is a unit vector in the direction of the joint axis
- $v_n = -\omega_n \times q_n$ where q_n is any point on the joint axis specified in the $\{s\}$ frame
- θ_n is the joint angle

Prismatic (sliding) joints

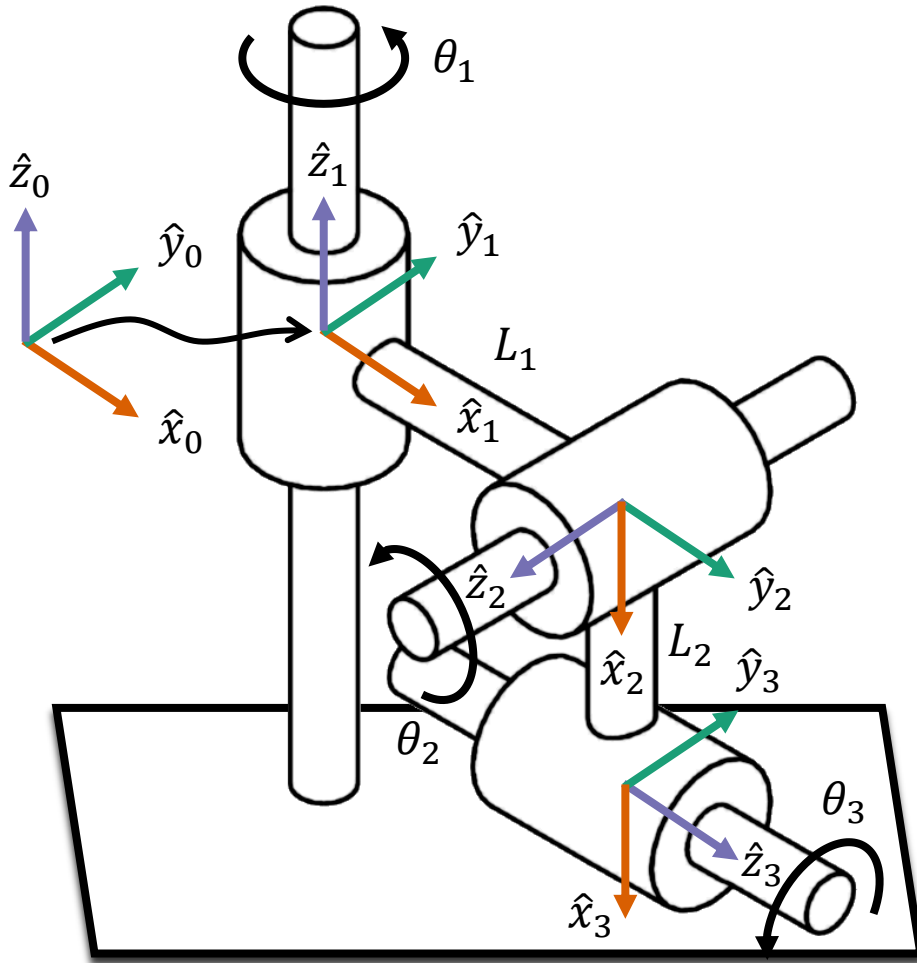
- $\omega_n = 0$
- $v_n \in \mathbb{R}^3$ is a unit vector in the direction of translation
- θ_n is the prismatic extension/retraction

*This is the **space form** of the Product of Exponentials formula, where screw axes are expressed in the $\{s\}$ frame*

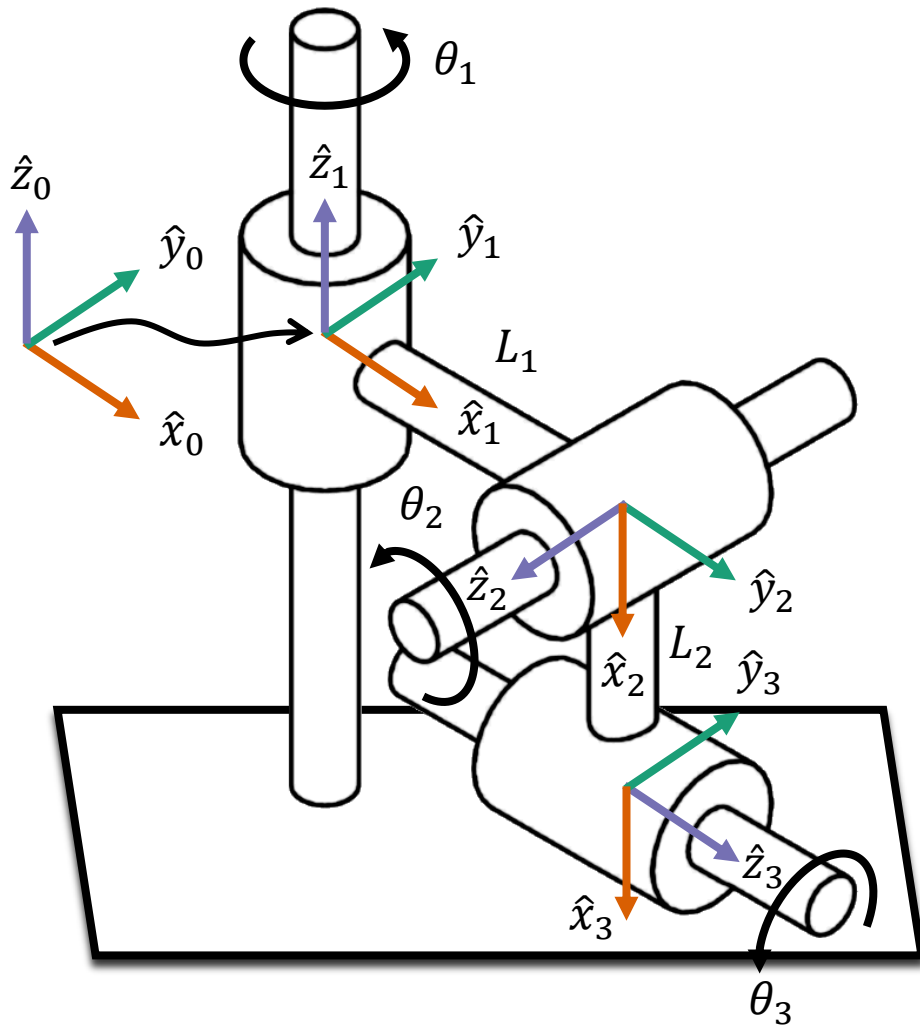
Product of Exponentials example: a 3R arm (1)

Consider the arm on the left with 3 revolute joints ($\theta_1, \theta_2, \theta_3$) and 2 links (L_1, L_2).

Let's use the PoE formula to solve the FK.



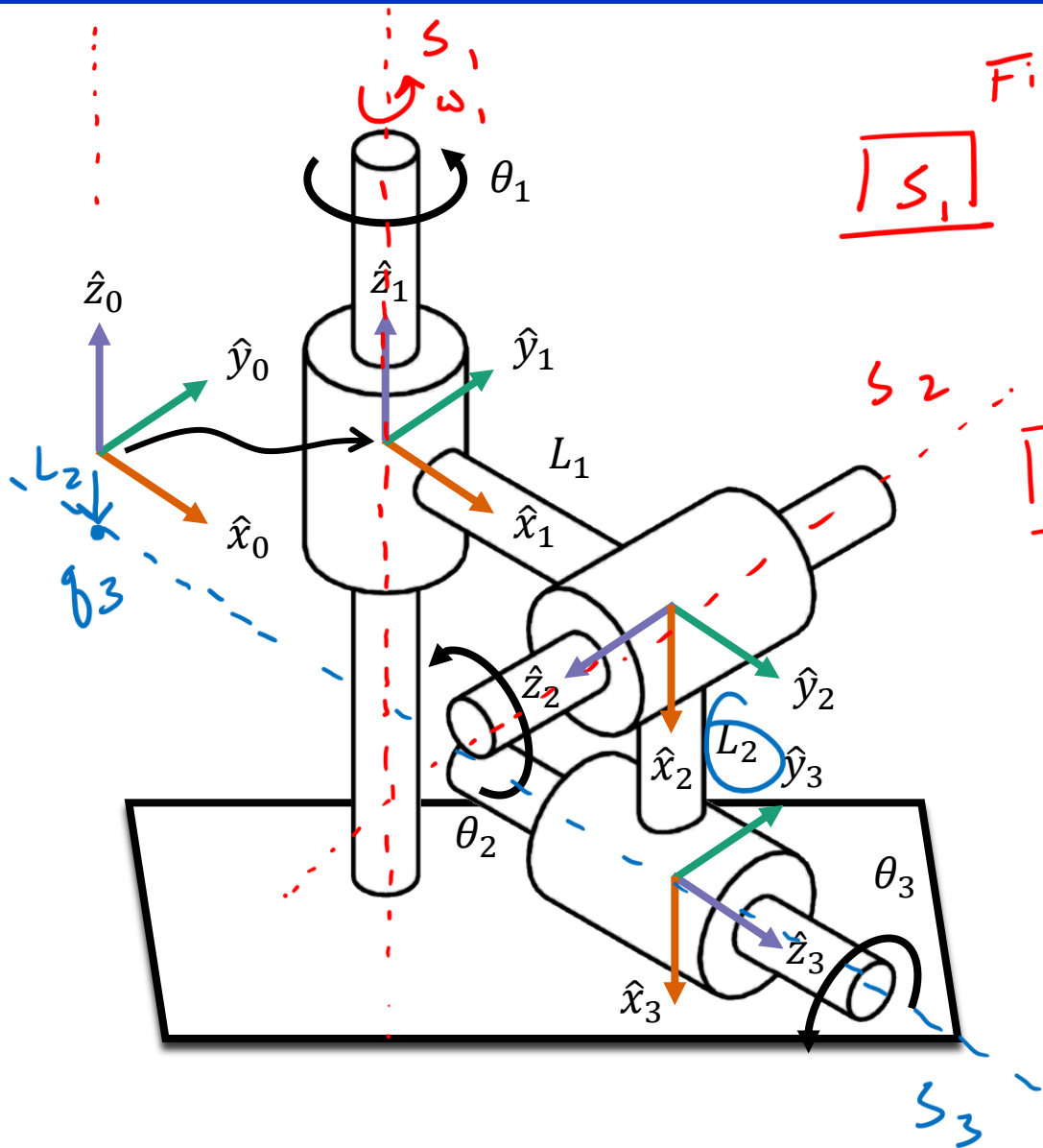
Product of Exponentials example: a 3R arm (2)



1. We can inspect the diagram to determine the home configuration:

$$M = \begin{bmatrix} 0 & 0 & 1 & L_1 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & -L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Product of Exponentials example: a 3R arm (3)



Find the screw axes:

$$\boxed{S_1}$$

$$\omega_1 = (0 \ 0 \ 1)$$

$$v_1 = (0 \ 0 \ 0)$$

$[q_1]$ is zero b/c S_1 passes through origin of $\{0\}$

$$\boxed{S_2}$$

$$\omega_2 = (0 \ -1 \ 0)$$

$$q_2 = (L_1 \ 0 \ 0)$$

$$v_2 = -\omega_2 \times q_2 = (0 \ 0 \ -L_1)$$

$$\boxed{S_3}$$

$$\omega_3 = (1 \ 0 \ 0)$$

$$q_3 = (0 \ 0 \ -L_2)$$

$$v_3 = -\omega_3 \times q_3 = (0 \ -L_2 \ 0)$$